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Methods and apparatus to provide welding power

Abstract

An example welding-type power supply includes: a transformer having a primary winding and first and second secondary windings; an input circuit configured to provide an input voltage to the primary winding of the transformer; first, second, third, and fourth switching elements, and a control circuit configured to: control the first, second, third, and fourth switching elements to selectively output a positive or negative output voltage without a separate rectifier stage by selectively controlling ones of the first, second, third, and fourth switching elements based on a commanded output voltage polarity and an input voltage polarity to the transformer; and prior to changing from a first output voltage polarity to a second output voltage polarity, controlling the first, second, third, and fourth switching elements to reverse the power flow to return reactive energy to an input circuit via the transformer.

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Background/Summary

(1) The present application claims the benefit of U.S. patent application Ser. No. 15/663,251, filed Jul. 28, 2017, entitled “METHODS AND APPARATUS TO PROVIDE WELDING POWER.” The entirety of U.S. patent application Ser. No. 15/663,251 is expressly incorporated herein by reference.

BACKGROUND

(2) This disclosure relates generally to welding systems and, more particularly, to methods and apparatus to provide welding power.

(3) In recent years, welding equipment has incorporated switched mode power supplies for converting and/or conditioning input power to welding power. Switched mode power supplies, or inverter-based power supplies, use semiconductor devices instead of more massive magnetic-based components, which substantially reduces the weight and size of the welding power supplies into which the inverter-based power supplies are implemented.

SUMMARY

(4) Methods and apparatus to provide welding power are disclosed, substantially as illustrated by and described in connection with at least one of the figures, as set forth more completely in the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a circuit diagram of a conventional full-bridge welding output circuit.
- (2) FIG. 2 is a graph illustrating example welding current outputs under different commutation schemes for the conventional full-bridge welding output circuit of FIG. 1.
- (3) FIG. 3 is a circuit diagram of a conventional half-bridge welding output circuit.
- (4) FIG. 4 is a circuit diagram of an example welding output circuit, in accordance with aspects of this disclosure.
- (5) FIG. 5 illustrates the operation of the example welding output circuit of FIG. 4, in which an output has an electrode-negative polarity and a positive input polarity.
- (6) FIG. 6 illustrates the operation of the example welding output circuit of FIG. 4, in which an output has an electrode-positive polarity and a positive input polarity.
- (7) FIG. 7 illustrates the operation of the example welding output circuit of FIG. 4, in which an output has an electrode-positive polarity and a negative input polarity.
- (8) FIG. 8 illustrates the operation of the example welding output circuit of FIG. 4, during a freewheeling phase in which there is no input power provided by a primary side inverter.
- (9) FIG. 9 is a graph of voltages and currents in the welding output circuit of FIG. 4.
- (10) FIG. 10 illustrates the operation of the example welding output circuit of FIG. 4, during a reverse power transfer from an output stage to an input stage.
- (11) FIG. 11 is a graph of voltages and currents in the welding output circuit of FIG. 4 during reverse power transfers as illustrated in FIG. 10.
- (12) FIG. 12 is a graph of current output by the welding output circuit of FIG. 4 during the reverse power transfer illustrated in FIG. 10.

- (13) FIG. 13 is a circuit diagram of the example welding output circuit of FIG. 4 including example heat sink assembly configurations to dissipate heat from switching elements.
- (14) FIG. 14 is a circuit diagram of an example welding output circuit using metal-oxide-semiconductor field effect transistors (MOSFETs), in accordance with aspects of this disclosure.
- (15) FIG. 15 is a circuit diagram of an example welding output circuit using multiple transformers, in accordance with aspects of this disclosure.
- (16) FIG. 16 is a block diagram of an example implementation of the control circuit, in which the control circuit is configured to output PWM signals to control the switching elements of FIGS. 4 and/or 14.

(17) The figures are not necessarily to scale. Where appropriate, similar or identical reference numbers are used to refer to similar or identical components.

DETAILED DESCRIPTION

- (18) Disclosed examples provide an alternating current (AC) current output from a switched mode power supply (SMPS)-based welding power source. In contrast with conventional full-bridge power supply circuits and half-bridge power supply circuits, disclosed example power supplies combine a rectifier stage with a commutation stage, using semiconductor devices, in a SMPS secondary circuit side of an isolation barrier.
- (19) Disclosed example methods and apparatus may be used to return reactive energy stored in the inductive weld circuit on the secondary side (e.g., output side) of the isolation barrier to the primary side (e.g., input circuit) of the isolation barrier by operating in a reverse manner.
- (20) As used herein, the term “welding-type power” refers to power suitable for welding, plasma cutting, induction heating, CAC-A and/or hot wire welding/preheating (including laser welding and laser cladding). As used herein, the term “welding-type power supply” refers to any device capable of, when power is applied thereto, supplying welding, plasma cutting, induction heating, CAC-A and/or hot wire welding/preheating (including laser welding and laser cladding) power, including but not limited to inverters, converters, resonant power supplies, quasi-resonant power supplies, and the like, as well as control circuitry and other ancillary circuitry associated therewith.
- (21) As used herein, the term “welding-type voltage” refers to a voltage suitable for welding, plasma cutting, induction heating, CAC-A, and/or hot wire welding/preheating (including laser welding and laser cladding).
- (22) Disclosed example welding-type power supplies include a transformer having first and second secondary windings, switching elements to control current flow from the first and second secondary windings to an output, and a control circuit configured to control the switching elements to selectively output a positive output voltage or a negative output voltage without a separate rectifier stage by selecting, based on an commanded output voltage polarity, a first subset of the switching elements to perform rectification.
- (23) In some examples, the control circuit selects a second subset of the switching elements based on the commanded output voltage polarity, and controls the second subset of the switching elements to couple the first and second secondary windings. In some such examples, the second subset is different than the first subset. In some examples, the control circuit selects the first subset from the switching elements and the second subset from the switching elements based on the input voltage polarity and the commanded output voltage polarity. In some examples, the control circuit re-selects the second subset of the switching elements when the commanded output voltage polarity changes.
- (24) Some example welding-type power supplies further include a primary converter circuit to convert input power to intermediate power having an intermediate frequency, where the transformer receives the intermediate power at a primary winding of the transformer. In some such examples, the control circuit controls the switching elements to supply energy to the primary converter circuit via the first and second secondary windings. Some examples further include a first heat sink to dissipate heat from a first set of at least two of the switching elements that share a first

electrical node, and a second heat sink to dissipate heat from a second set of at least two of the switching elements that share a second electrical node.

(25) In some examples, the transformer includes a primary winding, and the control circuit controls all of the switching elements to conduct current when substantially no voltage is applied to the primary winding of the transformer. In some examples, each of the switching elements includes an insulated gate bipolar transistor (IGBT) having a freewheeling diode or a metal oxide semiconductor field effect transistor (MOSFET).

(26) Disclosed welding-type power supplies include a transformer having first and second secondary windings, a first switching element, a second switching element, a third switching element, a fourth switching element, and a control circuit. The first switching element is coupled between the first secondary winding and a first output terminal of the welding-type power supply, the second switching element is coupled between the second secondary winding and the first output terminal of the welding-type power supply, the third switching element is coupled between the first secondary winding and a second output terminal of the welding-type power supply, and the fourth switching element is coupled between the second secondary winding and the second output terminal of the welding-type power supply. The control circuit controls the switching elements to output a welding voltage having a first polarity by controlling the first switching element and the second switching element to operate as a center tap between the first and second secondary windings while the third switching element and the fourth switching element operate as rectifiers. The control circuit further controls the switching elements to output the welding voltage having a second polarity by controlling the third switching element and the fourth switching element to operate as the center tap between the first and second secondary windings while the first switching element and the second switching element operate as rectifiers.

(27) In some examples, the first, second, third, and fourth switching elements include at least one of an insulated gate bipolar transistor (IGBT) having a freewheeling diode or a metal oxide semiconductor field effect transistor (MOSFET). In some examples, the control circuit controls a first one of the third switching element or the fourth switching element to conduct and to control the other one of the third switching element or the fourth switching element to be off based on a polarity of a primary winding voltage of the transformer. In some examples, the control circuit controls a first one of the first switching element or the second switching element to conduct and controls the other one of the first switching element or the second switching element to be off based on a polarity of a primary winding voltage of the transformer.

(28) Some example welding-type power supplies further include a primary converter circuit to convert input power to intermediate power having an intermediate frequency, in which the transformer receives the intermediate power at a primary winding of the transformer. In some such examples, the control circuit controls the first, second, third, and fourth switching elements to supply energy to the primary converter circuit via the first and second secondary windings.

(29) In some examples, the control circuit controls the first, second, third, and fourth switching elements to conduct current when substantially no voltage is applied to a primary winding of the transformer. In some examples, the first switching element, the first secondary winding, and the third switching element are coupled in series between the first output terminal and the second output terminal, and the second switching element, the second secondary winding, and the fourth switching element are coupled in series between the first output terminal and the second output terminal.

(30) In some examples, each of the first, second, third, and fourth switching devices conducts an average of half of an output current at the first and second output terminals. Some examples further include a first heat sink to dissipate heat from the first switching element and the second switching element, and a second heat sink configured to dissipate heat from the third switching element and the fourth switching element.

(31) FIG. 1 is a circuit diagram of a conventional full-bridge welding output circuit **100**. The

conventional full-bridge welding output circuit **100** includes a primary (input) side switched mode power supply **102**, which produces a high frequency (e.g., 18-100 kHz) signal to a primary winding **104** of a transformer **106**. The transformer **106** provides isolation between the primary side of the transformer (e.g., the primary power supply **102**) and a secondary side **108** of the transformer **106**.

(32) The secondary side **108** generates an output voltage and current, such as to a welding application. The secondary side **108** includes a rectifier stage **110**, a commutation stage **112**, and assist circuitry **114**. The rectifier stage **110** converts the high frequency signal to a DC current. The commutation stage **112** includes four transistors **116a-116d** arranged in a bridge to convert the DC current from the rectifier stage **110** to a lower frequency (e.g., 20-400 Hz) AC current (e.g., current suitable for welding). The assist circuitry **114** handles reactive energy present during commutation by clamping the output voltage and returning excess energy to the output.

(33) The secondary side **108** generates an output voltage between a welding-type electrode **118** (e.g., via a welding-type torch) and a workpiece **120**. A weld cable and/or the secondary side **108** may have an output inductance and/or a physical inductor, represented by an inductor **122**. Other sources of inductance, such as a coupling coil **124** for high frequency arc starting, may also provide reactive components.

(34) The conventional full-bridge topology of FIG. **1** has disadvantages, including excess losses caused by current continually flowing through three of the semiconductor devices in the secondary side **108** (e.g., one of the diodes in the rectifier stage **110** and two of the transistors **116a-116d** in the commutation stage **112**). Another disadvantage of the full-bridge topology is that only two of the four transistors **116a-116d** are used at any given time, while the other two of the transistors **116a-116d** (and the associated current paths) sit idle. Furthermore, the full-bridge topology requires the assist circuitry **114** to handle (e.g., clamp) the reactive energy stored in the output inductance.

(35) FIG. **2** is a graph illustrating example welding current outputs **202**, **204** under different commutation schemes for the conventional full-bridge welding output circuit **100** of FIG. **1**. One of the advantage of the full-bridge over other conventional topologies, such as the half-bridge topology, is that the full-bridge welding output circuit **100** is able to hard commute, or switch the current from one set of the transistors **116a-116d** to another set of the transistors **116a-116d**, and thereby change the direction of the current, with the use of clamping. The output current **202** illustrates an output using the hard commutation scheme and the output current **204** illustrates a soft commutation scheme using the assist circuitry **114**. Use of an output inductor **122** placed in the DC leg enables a more square output current as shown in the hard communication output current **202**. However, the hard commutation output current **202** results in a high change in current (e.g., dI/dt in the output inductor).

(36) FIG. **3** is a circuit diagram of a conventional half-bridge welding output circuit **300**. Like the full-bridge circuit **100** of FIG. **1**, the half-bridge welding output circuit **300** includes a primary (input) side switched mode power supply **302**, which produces a high frequency (e.g., 18-100 kHz) signal to a primary winding **304** of a transformer **306**. In the half-bridge welding output circuit **300**, the windings of the transformer **306** must be configured to be able to operate both in the first and third quadrants. The transformer **306** provides isolation between the primary side of the transformer (e.g., the primary power supply **302**) and a secondary side **308** of the transformer **306**.

(37) The half-bridge welding output circuit **300** includes two separate rectifier stages **310**, **312** to convert the high frequency signal from the transformer **306** to a DC current. The rectifier stage **310** performs rectification for EN polarity and the rectifier stage **312** performs rectification for EP polarity. The half-bridge welding output circuit **300** includes a commutation stage **314** including two transistors **316a**, **316b** to select between the two rectifier stages **310**, **312**. By alternating the selection between the rectifier stages **310**, **312**, the output current between a welding electrode **318** and a workpiece **320** is a lower frequency AC current (e.g., 20-400 Hz) suitable for welding.

(38) The half-bridge welding output circuit **300** includes assist circuitry **322** to clamp the voltage and return reactive energy present during commutation to the output.

(39) The advantages of the half-bridge welding output circuit **300** over the full-bridge welding output circuit **100** include eliminating one semiconductor device in the conduction path, which reduces heat loss. The half-bridge welding output circuit **300** also reduces part costs by replacing transistors with less expensive diodes.

(40) FIG. **4** is a circuit diagram of an example welding output circuit **400**. The example welding output circuit **400** is an improvement over the full-bridge and half-bridge circuits described above. In the half-bridge circuit **300**, two sets of rectifiers are present (e.g., one rectifier for EP current flow, the other for EN current flow). The transistors are then positioned to select between the sets of rectifiers based on the desired output polarity. In contrast, in the example welding output circuit **400**, the same semiconductor devices selectively perform rectification and commutation.

(41) As explained in more detail below, the welding output circuit **400** includes a transformer **402** having a primary winding **404**, a first secondary winding **406**, and a second secondary winding **408**. The welding output circuit **400** includes a set of switching elements **410a-410d**, and a control circuit **412** to control the switching elements. The welding output circuit **400** outputs AC and/or DC welding-type power via output terminals **414**, **416**. In the illustrated example, a first one of the output terminals **414** is coupled to a workpiece **418** (e.g., via a work cable) and the second one of the output terminals **416** is coupled to a welding electrode **420** (e.g., via a welding torch and a welding cable).

(42) The example switching elements **410a-410d** are insulated gate bipolar transistors (IGBTs) packaged with freewheeling diodes. However, in other examples other types of switching elements may be used. FIG. **14**, discussed below, illustrates an example using metal-oxide-semiconductor field effect transistors (MOSFETs) as the switching elements. The switching element **410a** is coupled between the secondary winding **406** and the output terminal **414** of the welding output circuit **400**. The switching element **410c** is coupled between the secondary winding **408** and the output terminal **414**. The switching element **410b** is coupled between the secondary winding **406** and the output terminal **416**. The switching element **410d** is coupled between the secondary winding **408** and the output terminal **416**. The switching element **410a**, the secondary winding **406**, and the switching element **410b** are coupled in series between the output terminal **414** and the output terminal **416**, and the switching element **410c**, the secondary winding **408**, and the switching element **410d** are coupled in series between the output terminal **414** and the output terminal **416**.

(43) Generally, the control circuit **412** controls the output voltage, output current, and/or output frequency by the welding output circuit **400**. To this end, the control circuit **412** controls the rectification and commutation functions by the switching elements **410a-410d** to cause the welding output circuit **400** to output welding voltages having a desired polarity (e.g., EP or EN). The control circuit **412** selects and controls a first subset of the switching elements **410a-410d** to function as a center tap between the secondary windings **406**, **408** based on the output polarity (e.g., EN or EP). The control circuit **412** selects and controls a second subset of the switching elements **410a-410d** to perform rectification by conducting current and/or blocking current based on an input polarity at the primary winding **404**. When the output polarity changes (e.g., EN to EP or EP to EN), the control circuit re-selects the first subset and the second subset from the switching elements **410a-410d**. Thus, in contrast with conventional output topologies, the control circuit **412**, the transformer **402**, and the switching elements **410a-410d** enable output of positive and negative output voltages and currents without a separate rectifier stage.

(44) As illustrated in FIG. **5**, the control circuit **412** outputs a welding voltage having a EN polarity by controlling the switching elements **410b**, **410d** to be constantly on (e.g., conducting) to operate as a center tap between the secondary windings **406**, **408**. The control circuit **412** controls the switching elements **410a**, **410c** to operate as rectifiers, which selectively conduct current based on the voltages across the secondary windings **406**, **408**.

(45) As illustrated in FIGS. **6** and **7**, the control circuit **412** outputs a welding voltage having a EP polarity by controlling the switching elements **410a**, **410c** to be constantly on (e.g., conducting) to

operate as the center tap between the secondary windings **406**, **408**. The control circuit **412** controls the switching elements **410b**, **410d** to operate as the rectifiers, which selectively conduct current based on the voltages across the secondary windings **406**, **408**.

(46) When one polarity is selected and the primary winding **404** is provided with input voltage, two of the switching elements **410a-410d** will be controlled to conduct, which configures the transformer **402** as a center tapped secondary with two output rectifiers. However, the orientation of the secondary windings **406**, **408** is configurable based on which two of the switching elements **410a-410d** are used to implement the center tap between the secondary windings **406**, **408**.

(47) Because the example switching elements **410a-410d** each have a diode component that can conduct current during rectification, the IGBT component of the switching elements **410a-410d** do not necessarily need to be controlled to conduct current when the switching element **410a-410d** is to permit current to flow during rectification. However, the control circuit **412** controls the IGBT component of the switching elements **410a-410d** to block current during rectification when the switching element **410a-410d** is not to conduct current. For example, in the EN output polarity and the input polarity illustrated in FIG. 5, the control circuit **412** may or may not control the IGBT of the switching element **410c** to conduct current (e.g., “turn on”), but controls the switching element **410a** to block current (e.g., “turn off”).

(48) In the welding output circuit **400**, the switching elements **410a-410d** implement two sets of rectifiers, and the control circuit **412** selects which set of rectifiers is to provide the output current. Each of the example switching elements **410a-410d** conducts an average of one-half of the output current.

(49) The welding output circuit **400** receives AC power at an intermediate frequency from a primary side inverter **422**. The primary side inverter **422** may generate the intermediate frequency from a primary power source, such as mains power and/or an engine-driven generator. While the transformer **402** is shown as one transformer, the welding output circuit **400** may include multiple transformers arranged in parallel or series. An example using multiple transformers is described below with reference to FIG. 15. The primary side inverter **422** and the welding output circuit **400** are capable of operating in any of the four quadrants of the V/A diagram (e.g., power supplying and power consuming quadrants).

(50) The welding output circuit **400** is illustrated with an output inductor **424** and a coupling coil **426**. The output inductor **424** and a coupling coil **426** may be similar or identical to the inductor **122** and the coupling coil **124** of FIG. 1.

(51) The example welding output circuit provides heat loss advantages similar to the conventional half-bridge topology of FIG. 3 (relative to the full-bridge and/or other topologies), where the output current flows through only two of the semiconductor devices. However, relative to the conventional half-bridge topology, which has two separate sets of rectifiers and only one of which is conducting at a given time, the example welding output circuit **400** has a higher utilization of all of the switching elements **410a-410d**.

(52) By combining the rectifier and current steering functions in the same semiconductor devices, the example welding output circuit **400** enables simplification of the routing of high current paths and/or improves thermal management by enabling grouping of the semiconductor devices into as few as two circuit locations from which heat is generated. For example, the heatsinks can be more effectively used (e.g., the duty cycle of heat sink heat dissipation can be increased (e.g., up to 100%)). In some such examples, the heatsinks may be configured such that areas of the heatsink(s) are always dissipating power, enabling to a more efficient use of real-estate, which may permit the design of more compact welding power sources.

(53) FIG. 8 illustrates the operation of the example welding output circuit **400** of FIG. 4, during a freewheeling phase in which there is no power provided by the primary side inverter **422**. The freewheeling phase may occur between positive voltage and negative voltage periods on the transformer primary **404** provided by the primary side inverter **422**. During the freewheeling phase,

stored energy in the secondary windings **406**, **408** and/or the output inductor **424** discharges to the welding output (e.g., to the arc). The current I_o is split between the two secondary windings **406**, **408**, so that each of the secondary windings conducts $I_o/2$. In the EP output polarity illustrated in FIG. **8**, the control circuit **412** controls the switching elements **410a** and **410c** to conduct. The control circuit **412** may also control the switching elements **410b** and **410d** to conduct via the transistor (e.g., synchronous rectification) and/or may permit the associated diodes of the switching elements **410b** and **410** to conduct the current.

(54) FIG. **9** is a graph of voltages and currents in the welding output circuit **400** of FIG. **4**. The graph illustrates a primary winding voltage **902** and a primary winding current **904** at the primary winding **404**. The graph further illustrates a first secondary current **906** (e.g., through the secondary winding **406** and the switching elements **410a**, **410b**) and a second secondary current **908** (e.g., through the secondary winding **408** and the switching elements **410c**, **410d**). The example primary winding voltage **902** and the currents **904**, **906**, **908** are illustrated for EP polarity operation of the example welding output circuit **400**. The freewheeling phase illustrated in FIG. **8** occurs during example time periods **910**, **912**.

(55) FIG. **10** illustrates the operation of the example welding output topology of FIG. **4**, during a reverse power transfer from the welding output circuit **400** to the primary inverter **422**. During the reverse power transfer mode, the welding output circuit provides power to the primary inverter **422** via the transformer **402**.

(56) In the conventional half-bridge topology discussed above, the current must be brought to zero before the current polarity can be reversed during commutation. Reduction of the current to zero is either done with additional circuitry, such as a clamp or other “assist” converter circuit, or by letting the current freewheel in the power source and decay naturally in the weld circuit. In contrast, the example welding output circuit may reverse the power flow to return reactive energy to the primary inverter **422**. Using the power reversal mode, the example control circuit **412** can more efficiently manage the decrease in current before polarity reversal.

(57) If, while current is flowing with the EP polarity, the control circuit **412** switches the output polarity selection to EN and controls the switching elements **410a-410d** according to the EN output polarity to return the power to the primary inverter **422** during an EP current polarity. The control circuit **412** controls the switching elements **410b** and **410d** to be on (e.g., conducting) and the switching elements **410a** and **410c** function as the rectifiers. The control circuit **412** controls the transistors of the switching elements **410a** and **410c** to be synchronous with the primary inverter **422**. Without the synchronous control, the welding output circuit **400** would not have a valid path for the current in the welding output circuit **400**. The control circuit **412** controls the switching element **410c** to be on when the voltage on the primary inverter **422** is positive, and controls the switching element **410a** to be on when the voltage on the primary inverter **422** is negative.

(58) In the operation shown in FIG. **10**, the control circuit **412** configures the secondary windings **406**, **408** of the transformer **402** as a current fed push-pull converter. The switching elements **410a**, **410c** function as source transistors feeding the push-pull converter. The primary inverter **422** includes switching devices that function as the rectifiers for the push-pull converter to return energy to, for example, a DC bus and/or an energy storage device that supplies power to the primary inverter **422** during normal operation (e.g., for providing power to the welding output).

(59) In a similar manner, the example control circuit **412** may control the switching elements **410a-410d** for an EP output polarity to return power when the current is flowing with an EN polarity.

(60) During the reverse power transfer operation, energy is being transferred from the output circuit at the secondary side of the transformer **402**, and the output current decreases. The voltage applied to the output circuit inductance (e.g., from the output terminal **416** to the electrode **420**, across the output inductor **424** and a coupling coil **426** is $V_{sub.arc} + V$ (e.g., the inverter voltage), where V_{arc} is the arc voltage between the electrode **420** and the workpiece **418**, and V is the voltage across the primary winding **404** of the transformer **402**. The secondary windings **406** and **408** also have the

voltage V as illustrated in FIG. 10. Without the reverse power transfer operation mode, the voltage applied to the output circuit inductance is $V_{\text{sub.arc}}$. The control circuit 412 may control the rate of current decrease by modulating a pulse width of the primary inverter 422 yielding an average voltage across the output inductor 424 and a coupling coil 426 of $V_{\text{sub.arc}} + \alpha V$, where α is the duty cycle. A wider pulse (e.g., a higher α) results in more reverse voltage, which removes more energy from the secondary circuit and decreases the output current faster.

(61) FIG. 11 is a graph of voltages and currents in the welding output topology of FIG. 4 during reverse power transfer operations as illustrated in FIG. 10. The graph of FIG. 11 illustrates a primary winding voltage 1102 and a primary winding current 1104 for the primary winding 404 of FIG. 10. The graph also illustrates a first secondary winding current 1106 through the secondary winding 408, and a second secondary winding current 1108 through the secondary winding 406.

(62) FIG. 12 is a graph of current output 1202 by the welding output circuit 400 of FIG. 4 during the reverse power transfer operation illustrated in FIG. 10. FIG. 12 also illustrates a current output 1204 using a natural current decay, where a reverse power transfer operation is not used and the welding output current is permitted to decay prior to commutation as in the conventional half-bridge topology. Comparing the current output 1202 using the reverse power transfer operation to the current output 1204 using natural decay, the use of the reverse power transfer operation permits a more rapid current decrease and, as a result, enables additional heat input by providing a higher current output prior to commutation.

(63) The use of the reverse power transfer mode provides significant advantages, including handling the reactive energy present in the welding output circuit 400 that must be removed from the welding output circuit 400 in order to commutate (e.g., reduce the current to OA before increasing in the opposite polarity). The energy is stored in an internal output inductor 424, in parasitic inductance of weld cables, and/or in other inductive elements. In conventional topologies, the current reduction is typically performed by a clamping capacitor connected to the weld circuit. The energy received at the clamping capacitor is then handled by a separate circuit, which returns the reactive energy to the input of the inverter, dissipates the energy with high power resistor(s), and/or recycles it to the weld circuit. In contrast, disclosed examples transfer the reactive energy toward the source of the input power (e.g., the primary side of the transformer 402, the primary inverter 422, and/or other connected circuitry) for handling by the main welding power source, without any additional circuitry.

(64) Another advantage of the reverse power transfer function is an increase in the voltage applied to the output circuit inductance (e.g., $V_{\text{sub.arc}} + \alpha V$). The increased voltage causes the current to be reduced faster than the rate of natural decay using only $V_{\text{sub.arc}}$ as the driving force. The increased voltage is particularly advantageous when operating on circuits with high inductance. In addition, the control circuit 412 can control the rate of current decline by modulating the duty cycle α . Conventionally, the current decline would either be rapid decay (i.e., a hard commutation by dumping the energy into a high voltage clamp) or a slow natural decay. Waiting for the current to decay naturally reduces the available heat input to the welding output, because more time is spent at a current lower than is desired. By controlling the slope of the current, the control circuit 412 may improve arc stability and/or lower audio emissions while maintaining heat input.

(65) Using disclosed example circuits in the reverse power transfer mode enables an output current that more closely resembles a square wave (e.g., more like the hard commutation output current 202 in FIG. 2) while also controlling and limiting dI/dt .

(66) FIG. 13 is a circuit diagram of the example welding output circuit 400 of FIG. 4 including example heatsink assemblies 1302, 1304 to dissipate heat from the switching elements 410a-410d. The example switching elements 410a-410d may be discrete semiconductor devices (e.g., TO-247 package, TO-220 package, etc.) and/or semiconductor devices with packages having non-isolated base plates. In such examples, the heat sink tabs of the switching elements 410a-410d connect to one of only two nodes. Thus, the example heatsink assemblies 1302, 1304 can provide heat

dissipation and cooling for the entire welding output circuit **400** with only two separate, non-isolated heatsinks.

(67) Conventional topologies require at least three heat sink assemblies. In contrast, the example welding output circuit **400** do not isolate the heatsink assemblies **1302**, **1304** from the switching elements **410a-410d**, which enables a reduction in the number of heatsinks. The reduction in heatsinks may reduce cost and/or improve thermal performance of the welding output circuit **400** by removing at least one barrier to heat leaving the switching elements **410a-410d** (relative to conventional topologies) and removing at least one component that could possibly fail under thermomechanical stress cycling. Additionally, the heatsink assemblies **1302**, **1304** can function as a part of the current conducting path, thereby simplifying electrical routing of the circuit(s) and/or circuit boards.

(68) In contrast with conventional topologies which concentrate current through particular switching devices while other devices are relatively idle, disclosed example welding output circuits balance current through the paths because the switching elements **410a-410d** are used either as rectifiers or switches at any given time and, thus, balance heat dissipation between the switching elements **410a-410d**. As a result, heat dissipation by the heatsink assemblies **1302**, **1304** is more evenly distributed relative to conventional topologies, which leave some devices sitting idle and corresponding areas of the heatsink sitting cool, while other device and heatsink areas are hotter. Because the heatsink assemblies **1302** and **1304** dissipate approximately the same amount of heat, regardless of the output polarity, the two heatsink assemblies **1302**, **1304** can be designed to be similar or identical.

(69) FIG. **14** is a circuit diagram of an example welding output circuit **1400** using metal-oxide-semiconductor field effect transistors (MOSFETs). The example welding output circuit **1400** is similar to the welding output circuit **400** of FIG. **4**, and includes the transformer **402** (e.g., the primary winding **404** and the secondary windings **406**, **408**), the control circuit **412**, the terminals **414**, **416**, the primary inverter **422**, the output inductor **424**, and the coupling coil **426**. The example welding output circuit **1400** includes switching elements **1402a-1402d** (e.g., corresponding to the switching elements **410a-410d**). The example switching elements **1402a-1402d** each include a MOSFET transistor. Because MOSFET transistors have intrinsic diodes in parallel with the transistor, use of the switching elements **1402a-1402d** including MOSFET transistors enables the use of the intrinsic diodes to perform rectification (e.g., as rectifier diodes).

(70) When the channel of the MOSFET is on, the MOSFET conducts current in either direction. With low voltage components, the resistance of the channel during conduction can be significantly lower than the voltage drop of the PN junction of traditional fast diode rectifiers. The example control circuit **412** may control the switching elements **1402a-1402d** selected for rectification using synchronous rectification to reduce the power consumption in the switching elements **1402a-1402d** performing rectification. For example, for the EP output polarity illustrated in FIGS. **6** and **7**, while the MOSFETs of the switching elements **1402b**, **1402d** may be controlled to be off (e.g., blocking or non-conducting) to let the associated intrinsic diodes conduct, the control circuit **412** may turn the MOSFETs of the switching elements **1402b**, **1402d** on (e.g., conducting) synchronously with the polarity of the voltage on the primary winding **404**. When the voltage applied to the primary winding **404** is positive (e.g., FIG. **6**), the control circuit **412** may synchronously turn the IGBT of the switching element **1402b** on and turn the IGBT of the switching element **1402d** off. Conversely, when the voltage applied to the primary winding **404** is negative (e.g., FIG. **7**), the control circuit **412** may synchronously turn the IGBT of the switching element **1402d** on and turn the IGBT of the switching element **1402b** off. Therefore, using synchronous rectification with the MOSFETs coupled with low voltage MOSFETs could further reduce power loss in the rectifier function. By using MOSFET transistors, the number of nodes, current paths, and/or high current interconnections in the welding output circuit **1400** is reduced (relative to the welding output circuit **400** of FIG. **4**), which simplifies component layout and/or reduces cost.

(71) In some examples, the welding output circuit may include a combination of MOSFETs and IGBTs to implement the switching elements.

(72) FIG. 15 is a circuit diagram of an example welding output circuit 1500 using multiple transformers. When using discrete devices, an advantageous option is to use two or more transformers 1502, 1504. The transformer 1502 has a primary winding 1506 and secondary windings 1508, 1510. The transformer 1504 has a primary winding 1512 and secondary windings 1514, 1516. The primary windings 1506 and 1512 are placed in series, and the secondary windings 1508, 1510, 1514, and 1516 are in parallel.

(73) The welding output circuit 1500 includes switching elements 1518a-1518h to couple the secondary windings 1508, 1510, 1514, and 1516 to output terminals 1520, 1522. The example switching elements 1518a-1518h are implemented using MOSFETs. A control circuit 1524 controls the switching elements 1518a-1518h to conduct and/or to block current. However, a PN junction in each of the switching elements 1518a-1518h may conduct current while the switching element is controlled to be off or non-conducting. The example welding output circuit 1500 includes an output inductor 1526 and a coupling coil 1528.

(74) Having the primary windings 1506, 1512 of the transformers 1502, 1504 in series causes the transformers 1502, 1504 to each conduct substantially the same amount of current and/or dissipate substantially the same heat load, thereby improving reliability. The forced current sharing also causes the current flowing through the secondary side commutator and rectifier switching elements 1518a-1518h to be split substantially equally. When building the welding output circuit 1500 with discrete devices (e.g., TO-220 packages, TO-247 packages, etc.), switching elements may resist sharing the load current equally (e.g., due to variations in parameter, unequal cooling, etc.).

(75) In some examples, turns ratio(s) of the transformers 1502, 1504 can be lower than the turns ratio using a single transformer (e.g., as in FIG. 4), because as the voltage is already split once on between the primary windings 1506, 1512. For example, in place of an 4:1 turns ratio with a single transformer, two transformers in a series configuration could use a 2:1 turns ratio. A lower turns ratio can simplify the transformer design and/or improve parameters such as leakage inductance.

(76) While FIG. 15 illustrates two transformers 1502, 1504 in series, other examples may include more transformers to share current between even more paths.

(77) FIG. 16 is a block diagram of an example implementation of the control circuit 412 in which the control circuit 412 is configured to output PWM signals to control the switching elements 410a-410d and/or 1402a-1402d of FIGS. 4 and/or 14.

(78) The example controller 412 of FIG. 16 may be used to control AC and DC welding outputs, such as pulsed DC applications (EN and/or EP polarity). The ability to rapidly reduce the current in the weld circuit (as discussed above with reference to FIG. 13) and/or controlling the slope of the current provides benefits even when DC welding and the polarity of the current does not need to be reversed. For example, the example control circuit 412 may improve pulsed DC application (EN or EP) by controlling the output current to more closely match an ideal square pulse train (e.g., a pulse train having zero current rise time and/or zero current fall time). Additionally or alternatively, in GMAW, after a short circuit condition has cleared and the output current is high, the control circuit 412 can now return more rapidly to the lower current level required for the arc. In conventional DC topologies, the only forces reducing the current are resistive losses in the welding power supply, the welding circuit, and the arc. As explained above, the control circuit 412 may control the switching elements 410a-410d and/or 1402a-1402d in reverse mode as described above to add an additional modulated voltage to reduce the output current more rapidly.

(79) By selecting the appropriate switching elements 410a-410d and/or 1402a-1402d, the control circuit 412 may operate either in direct or reverse power mode and effectively apply a positive or negative voltage source to the output. Conventional welding processes regulate the output current to a desired commanded current. As shown in FIG. 16, the control circuit 412 receives a desired current command 1602 and a current feedback signal 1604. The control circuit 412 compares the

current command **1602** to the current feedback signal **1604** to generate an error signal **1606**. Both of the current command **1602** and the current feedback signal **1604** signals include the respective signal polarities, and the error signal **1606** also contains polarity information.

(80) The control circuit **412** includes a dynamic controller **1608** to, based on the error signal **1606**, determine both an amplitude **1610** and a polarity **1612** of the voltage source to be applied to the weld output to achieve the commanded current. The example control circuit **412** may be implemented by an analog proportional-integral-derivative (PID) controller, though other analogue and/or digital control schemes may be used. The amplitude **1610** and polarity **1612** signals are then provided to a pulse width modulator (PWM) circuit **1614** to generate gate commands **1616a-1616d** with the appropriate pulse widths to control the switching elements **410a-410d** and/or **1402a-1402d**. The PWM circuit **1614** may further generate PWM signals **1616e**, **1616f** to control one or more switching elements in the primary inverter **422**. The PWM circuit **1614** may be a voltage mode control type or a current mode control type. Working with the PWM signals allows the control circuit **418** full dynamic control (e.g., amplitude and polarity) of the source applied to the output and, therefore, can yield improved transient results.

(81) The present methods and systems may be realized in hardware, software, and/or a combination of hardware and software. The present methods and/or systems may be realized in a centralized fashion in at least one computing system, or in a distributed fashion where different elements are spread across several interconnected computing systems. Any kind of computing system or other apparatus adapted for carrying out the methods described herein is suited. A typical combination of hardware and software may include a general-purpose computing system with a program or other code that, when being loaded and executed, controls the computing system such that it carries out the methods described herein. Another typical implementation may comprise an application specific integrated circuit or chip. Some implementations may comprise a non-transitory machine-readable (e.g., computer readable) medium (e.g., FLASH drive, optical disk, magnetic storage disk, or the like) having stored thereon one or more lines of code executable by a machine, thereby causing the machine to perform processes as described herein. As used herein, the term “non-transitory machine-readable medium” is defined to include all types of machine readable storage media and to exclude propagating signals.

(82) As utilized herein the terms “circuits” and “circuitry” refer to physical electronic components, any analog and/or digital components, power and/or control elements, such as a microprocessor or digital signal processor (DSP), or the like, including discrete and/or integrated components, or portions and/or combination thereof (i.e. hardware) and any software and/or firmware (“code”) which may configure the hardware, be executed by the hardware, and or otherwise be associated with the hardware. As used herein, for example, a particular processor and memory may comprise a first “circuit” when executing a first one or more lines of code and may comprise a second “circuit” when executing a second one or more lines of code. As utilized herein, “and/or” means any one or more of the items in the list joined by “and/or”. As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. In other words, “x and/or y” means “one or both of x and y”. As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, “x, y and/or z” means “one or more of x, y and z”. As utilized herein, the term “exemplary” means serving as a non-limiting example, instance, or illustration. As utilized herein, the terms “e.g.,” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. As utilized herein, circuitry is “operable” to perform a function whenever the circuitry comprises the necessary hardware and code (if any is necessary) to perform the function, regardless of whether performance of the function is disabled or not enabled (e.g., by a user-configurable setting, factory trim, etc.).

(83) While the present method and/or system has been described with reference to certain implementations, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present method and/or

system. For example, block and/or components of disclosed examples may be combined, divided, re-arranged, and/or otherwise modified. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. Therefore, the present method and/or system are not limited to the particular implementations disclosed. Instead, the present method and/or system will include all implementations falling within the scope of the appended claims, both literally and under the doctrine of equivalents.

Claims

1. A welding-type power supply, comprising: a transformer having first and second secondary windings; switching elements configured to control current flow from the first and second secondary windings to an output; and a control circuit configured to control the switching elements to selectively output a positive output voltage or a negative output voltage without a separate rectifier stage by: selecting, based on a commanded output voltage polarity, a first subset of the switching elements to couple the first and second secondary windings, wherein the selection of the first subset of the switching elements remains the same while the commanded output voltage polarity remains the same; and selecting, based on an input voltage polarity, a second subset of the switching elements to perform rectification, wherein the second subset of the switching elements is different than the first subset of the switching elements.
 2. The welding-type power supply as defined in claim 1, wherein the control circuit is configured to re-select the first subset of the switching elements when the commanded output voltage polarity changes.
 3. The welding-type power supply as defined in claim 1, further comprising a primary converter circuit configured to convert input power to intermediate power having an intermediate frequency, the transformer configured to receive the intermediate power at a primary winding of the transformer.
 4. The welding-type power supply as defined in claim 3, wherein the control circuit is configured to control the switching elements to supply energy to the primary converter circuit via the first and second secondary windings.
 5. The welding-type power supply as defined in claim 1, further comprising: a first heat sink configured to dissipate heat from a first set of at least two of the switching elements that share a first electrical node; and a second heat sink configured to dissipate heat from a second set of at least two of the switching elements that share a second electrical node.
 6. The welding-type power supply as defined in claim 1, wherein the transformer comprises a primary winding, the control circuit configured to control all of the switching elements to conduct current when substantially no voltage is applied to the primary winding of the transformer.
 7. The welding-type power supply as defined in claim 1, wherein each of the switching elements comprises an insulated gate bipolar transistor (IGBT) having a freewheeling diode or a metal oxide semiconductor field effect transistor (MOSFET).
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